

COMPUTER PROGRAMMING FOR COMPUTER ENGINEERS

INDEX NUMBER : 4072924

PROJECT 4- Data Analysis & Visualisation

1. Data Loading and Cleaning

The dataset (well_data.csv) was loaded using Pandas. Three derived columns were added:

$GOR \text{ (scf/STB)} = \text{Gas Rate (Mscf/day)} \times 1000 / \text{Oil Rate (STB/day)}$

$WOR = \text{Water Rate (STB/day)} / \text{Oil Rate (STB/day)}$

$\text{Liquid Rate (STB/day)} = \text{Oil Rate} + \text{Water Rate}$

MON TH	OIL_ST BDAY	GAS_MSC FDAY	WATER_STBDAY	BHP_PSI	WHP PSI	GOR	WO R	Liquid_ST Bday
1	1200	850	60	3400	280	708.33	0.050	1260
2	1050	790	90	3250	260	752.38	0.086	1140
3	920	730	125	3100	245	793.48	0.136	1045
4	810	680	165	2950	235	839.51	0.204	975
5	710	625	200	2810	225	880.28	0.282	910
6	630	575	240	2680	215	912.70	0.381	870
7	555	532	285	2560	207	958.56	0.514	840
8	490	492	335	2450	200	1004.08	0.684	825
9	430	455	390	2350	194	1058.14	0.907	820
10	380	421	450	2260	188	1107.89	1.184	830
11	335	389	515	2180	183	1161.19	1.537	850
12	295	360	585	2110	178	1220.34	1.983	880

2. Summary Statistics

Descriptive statistics (mean, standard deviation, min, max, quartiles) were generated for all variables using df.describe(). Key findings:

Month of peak oil production: Month 1, at 1200 STB/day

Month WOR first exceeds 0.25: Month 5 (WOR = 0.282)

Total 12-month cumulative oil production: 237.56 MSTB

Cumulative oil was calculated by multiplying each month's oil rate by the average number of days per month (30.4375), summing over all 12 months, and converting to thousand stock-tank barrels (MSTB).

3. Visualisation

A 4-panel dashboard was generated using Matplotlib:

- (a) Oil rate vs month — bar chart, showing the steady production decline
- (b) GOR vs month — line plot, showing a rising trend consistent with reservoir pressure depletion below the bubble point
- (c) BHP vs month — scatter plot with a linear trendline, confirming a consistent pressure decline
- (d) Oil rate vs water rate crossplot, illustrating the inverse relationship as water cut increases over the well's life

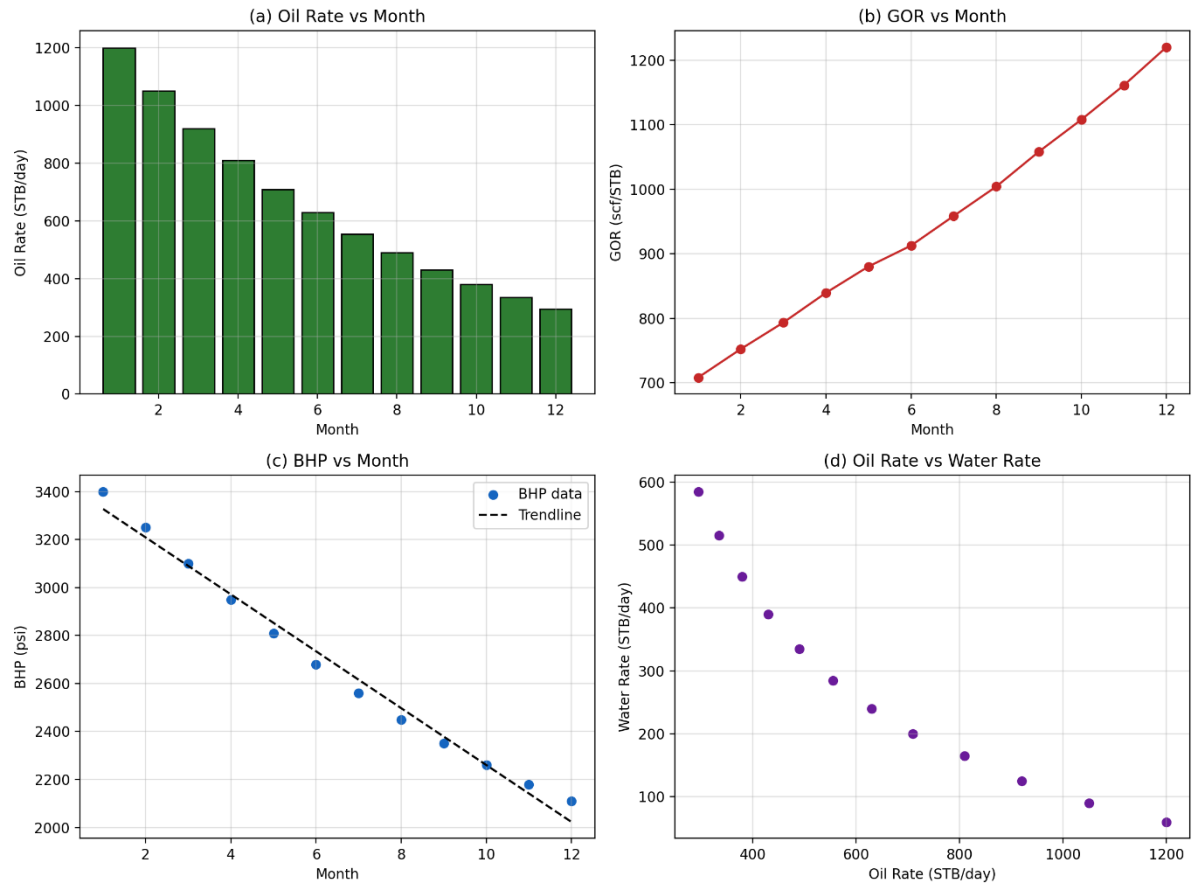


Figure 1: Well Performance Dashboard — Oil Rate, GOR, BHP, and Water Rate Trends (Months 1–12).

4. Exponential Decline Curve Analysis

The exponential decline model $q = q_i \cdot e^{(-Dt)}$ was linearised to $\ln(q) = \ln(q_i) - D \cdot t$ and fitted using NumPy's polyfit function, with t expressed in days (month number $\times 30.4375$) so that D represents a standard nominal decline rate.

Results of the fit:

Initial rate, $q_i = 1350.7$ STB/day

Nominal decline rate, $D = 0.004171$ /day (1.524 /year)

Economic limit ($q_{ab} = 100$ STB/day) reached at $t \approx 624$ days (approximately 20.5 months)

Estimated Ultimate Recovery (EUR) at $q_{ab} = 100$ STB/day = 299.84 MSTB

EUR was calculated analytically from the closed-form integral of the exponential decline equation:

$$EUR = (q_i - q_{ab}) / D.$$

5. Appendix — Python Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("well_data.csv")
df["GOR"] = df["GAS_MSCFDAY"] * 1000 / df["OIL_STBDAY"]
df["WOR"] = df["WATER_STBDAY"] / df["OIL_STBDAY"]
df["Liquid_STBday"] = df["OIL_STBDAY"] + df["WATER_STBDAY"]

print(df.describe())

peak_month = int(df.loc[df["OIL_STBDAY"].idxmax(), "MONTH"])
wor_month = int(df[df["WOR"] > 0.25]["MONTH"].iloc[0])
days_per_month = 365.25 / 12
cum_oil_mstb = (df["OIL_STBDAY"] * days_per_month).sum() / 1000

t_days = df["MONTH"].values * days_per_month
slope, intercept = np.polyfit(t_days, np.log(df["OIL_STBDAY"].values), 1)
D = -slope
```

```

qi = np.exp(intercept)
q_ab = 100.0
t_ab_days = np.log(qi / q_ab) / D
EUR_mstb = (qi - q_ab) / D / 1000

fig, axs = plt.subplots(2, 2, figsize=(12, 9))
axs[0,0].bar(df["MONTH"], df["OIL_STBDAY"], color="green")
axs[0,0].set_title("(a) Oil Rate vs Month")
axs[0,1].plot(df["MONTH"], df["GOR"], marker="o", color="red")
axs[0,1].set_title("(b) GOR vs Month")
axs[1,0].scatter(df["MONTH"], df["BHP_PSI"], color="blue")
z = np.polyfit(df["MONTH"], df["BHP_PSI"], 1)
axs[1,0].plot(df["MONTH"], np.poly1d(z)(df["MONTH"]), "--", color="black")
axs[1,0].set_title("(c) BHP vs Month")
axs[1,1].scatter(df["OIL_STBDAY"], df["WATER_STBDAY"], color="purple")
axs[1,1].set_title("(d) Oil Rate vs Water Rate")
plt.tight_layout()
plt.savefig("well_performance_dashboard.png", dpi=200)

```